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Indicile Gini

Indicele Gini

- Daca T este o multime de date ce contine inreg din n clase (deci atributul tinta are n valori discrete posibile), indicele Gini se defineste:

$$gini(T) = 1 - \sum_{j=1}^n p_j^2$$

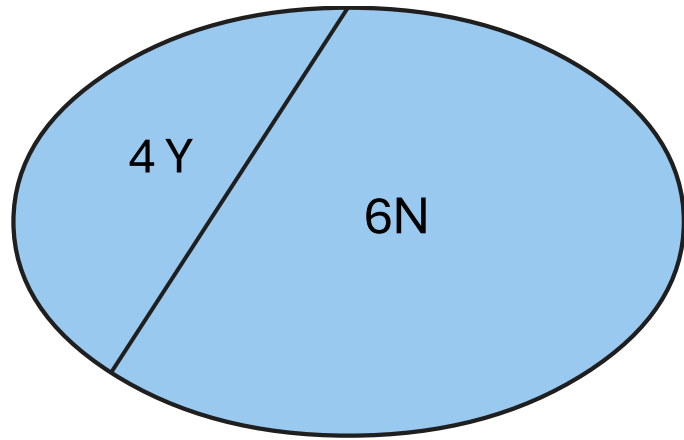
unde p_j = frecventa relativa a clasei j in T

$$p_j = \frac{nr\ aparitii\ ale\ clasei\ j}{nr\ inreg\ ale\ colectiei\ T}$$

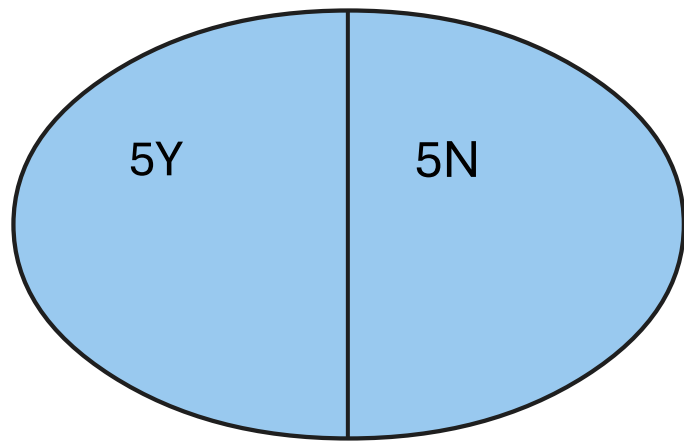
- Daca T se imparte in submultimile astfel incat T are N elemente, T₁ are N₁ elemente, ..., T_k are N_k elemente, atunci indicele Gini al diviziunii

$$gini_{divizune}(T) = \frac{N_1}{N} gini(T_1) + \frac{N_2}{N} gini(T_2) + \dots + \frac{N_k}{N} gini(T_k)$$

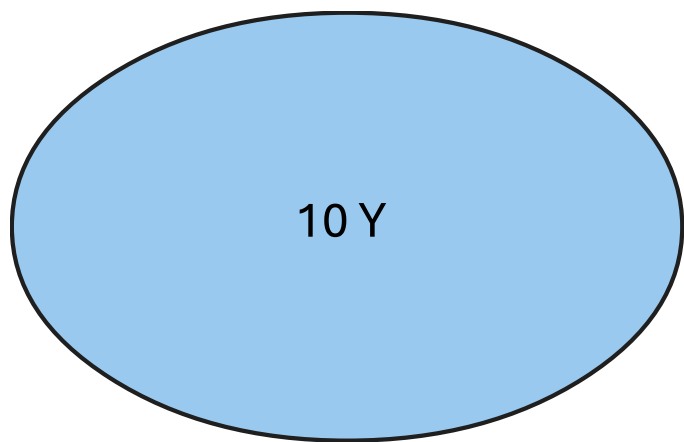
	workclass	relationship	income
0	State-gov	Not-in-family	<=50K
1	Self-emp-not-inc	Husband	<=50K
2	Private	Not-in-family	<=50K
3	Private	Husband	<=50K
4	Private	Wife	<=50K
5	Private	Wife	<=50K
6	Private	Not-in-family	<=50K
7	Self-emp-not-inc	Husband	>50K
8	Private	Not-in-family	>50K
9	Private	Husband	>50K



$gini(T) = 1 - p_Y^2 - p_N^2 = 1 - (4/10)^2 - (6/10)^2 = 1 - 16/100 - 36/100 = 1 - 52/100 = 48/100 = 0.48$



$gini(T) = 1 - p_Y^2 - p_N^2 = 1 - (5/10)^2 - (5/10)^2 = 1 - (1/2)^2 - (1/2)^2 = 1 - 1/4 - 1/4 = 1 - 1/2 = 1/2 = 0.5$



- Indicele Gini ne da probabilitatea ca doua elemente alese la intamplare din colectia data sa nu fie in aceeasi clasa.

Intrucat indicele Gini a fost calculat pentru diviziunea dupa workclass, procedam la calcularea indicelui Gini dupa 'relationship'.

Relationship	Income	<=50k	> 50k
Not-in-family		3	1
Husband		2	2
Wife		2	0

Notam multimea 'Not-in-family' cu T1:

$gini(T1) = 1 - (3/4)^2 - (1/4)^2 = 1 - 9/16 - 1/16 = (16-9-1) / 16 = 6 / 16 = 0.375$

Notam multimea 'Husband' cu T2:

$gini(T2) = 1 - (2/4)^2 - (2/4)^2 = 1 - 1/4 - 1/4 = (4-1-1) / 4 = 2/4 = 0.5$

Notam multimea 'Wife' cu T3:

$gini(T3) = 1 - (2/2)^2 - (0 / 2)^2 = 1 - 1 = 0$

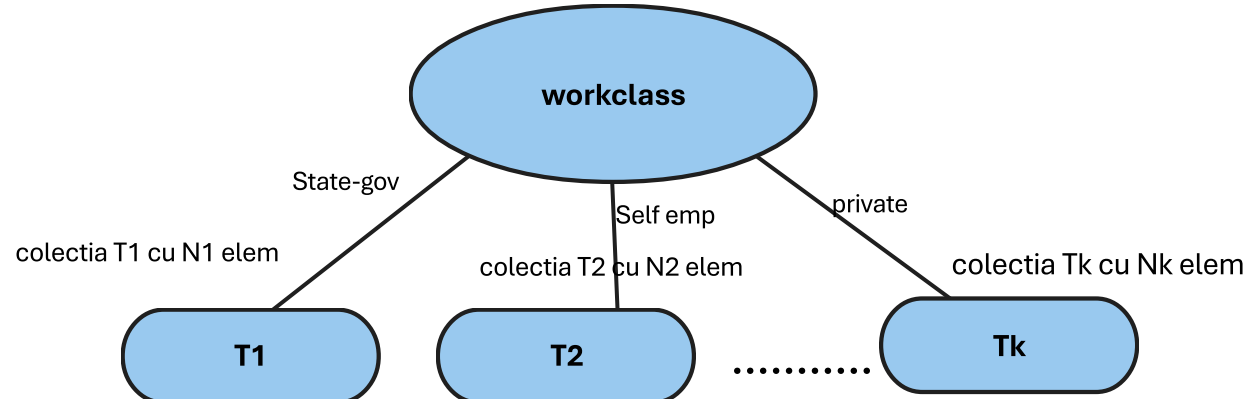
$gini_{diviz_relationship} = 4/10 * gini(T1) + 4/10 * gini(T2) + 2/10 * gini(T3) = 4/10 * 0.375 + 4/10 * 0.5 + 2/10 * 0 = 0.15 + 0.2 + 0 = 0.35$

Daca gini pentru diviziunea workclass este 0.4286 (din whiteboard curs) iar gini pentru diviziunea relationship este 0.35, vom alege valoarea mai mica, respectiv diviziunea relationship. In whiteboard curs a fost facuta o gresala, n fiind 10 nu 9 (sunt 10 elemente), ceea ce rezulta un gini de 0.38 (tot mai mare decat 0.35). In ambele situatii, vom alege diviziunea relationship cu gini = 0.35.

Sa se determine cel mai bun atribut pentru diviziune folosind criteriul Gini

income este atributul tinta cu valorile >50 K , <=50K

1. Calculam indicele Gini diviziune dupa workclass



workclass \ income	<=50K	>50K	
state_gov	1	0	T1
self emp	1	1	T2
private	5	2	T3

$gini(T1) = 0$

$gini(t2)= 0.5$

$gini(t3)= 1- (5/7)^2 - (2/7)^2 = 1-(25+4)/49 = 1-29/49=20/49$

$dini_diviz_workclass = (1/10) * gini(t1) + (2/10) * gini(t2) + (7/10) * gini(t3) = (1/10) * 0 + (2/10) * 0.5 + (7/10) * (20/49) = 27/70 = 0.3857$

I .Sa se continue exercitiul cu aflarea celui mai bun atribut

II. Sa se identifice cel mai bun atribut din arbore de decizie

urmator si valoarea corespunzatoare pe ramuri.

Cele mai bune atribute sunt:

1. La nivelul 2 de sus in jos, vom alege atributul 'fare <= 26.269' (unde gini 0.316 < 0.38);
2. La nivelul 3 de sus in jos, vom alege atributul 'age <= 13.5' (unde gini 0.232 < 0.457);
3. La nivelul 4 de sus in jos, vom alege atributul 'embarked_C <= 0.5' (unde gini 0.188 < 0.26);
4. La nivelul 5 de sus in jos, vom alege atributul 'fare <= 13.25' (unde gini 0.166 < 0.332);

